Deep Belief Network Enabled Surrogate Modeling for Fast Preventive Control of Power System Transient Stability

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***Abstract*—The widely used transient stability-constrained optimal power flow (TSC-OPF) method for power system preventive control is very time-consuming and thus not applicable for large-scale systems. This article proposes a new deep learning-enabled surrogate model that can significantly improve computational efficiency while maintaining high accuracy. To achieve that, the deep belief network (DBN) is strategically integrated with the reference-point-based nondominated sorting genetic algorithm (NSGA-III) to develop a new preventive control framework. The DBN allows us to identify the mapping relationship between the transient stability index and system operational features. The identified functional mapping relationship is further used as the surrogate to connect the DBN results with TSC-OPF for preventive control. The integrated NSGA-III and surrogate model enable the multiobjective optimization to consider various constraints and objectives, such as minimization of costs of generation dispatch cost and load shedding while maintaining the system stability. Extensive simulation results on several IEEE test systems show that the proposed method can achieve highly efficient control solutions and outperform other alternatives in terms of computational efficiency and economic benefits.**

***Index Terms*—Deep belief network (DBN), deep learning, nondominated sorting genetic algorithm (NSGA-III), surrogate model, transient stability preventive control.**

I. Introduction

WITH the increased penetration of renewable energy and scale of the power system, the system is operated more and more close to their stability limits. Therefore, the development of effective preventive controls for transient stability is of vital importance [1].

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The traditional transient stability assessment usually relies on time-domain simulations. However, that is computationally intensive for practical large-scale systems [2]. To deal with that, several other methods have been proposed, including the phase-plane trajectories [3], the Lyapunov functions family [4], the ensemble Kalman filter [5], the unscented transformation [6], the transient energy function [7], and the nonparametric model [8]. Machine learning-based tools have also been applied for transient stability prediction. In [9], the improved decision tree is developed for online dynamic security assessment using synchrophasor measurements. It considers all possible paths instead of the classification results at each terminal node. Additionally, the Lasso regression is used to predict the transient stability boundary [10]. In [11], bus voltage magnitudes, the generator rotor speeds, and rotor angles are utilized as input parameters for the support vector machine (SVM) to predict transient stability. Instead of SVM, the artificial neural network (ANN) [12], [13] is used for that purpose as well. However, these traditional machine learning algorithms have rather limited capabilities of dealing with system nonlinearity and generalization. To overcome these issues, deep learning models are developed and applied in many fields. Compared to the traditional machine learning techniques, deep learning has the advantages of automatic feature extraction, strong abstraction ability, and good convergence. Particularly, the deep neural network structure has a larger number of neural layers. This allows uncovering the underlying knowledge contained in a large set of data. In [14], for instance, the stacked denoising autoencoder is used for transient stability prediction. However, it does not answer the question of how to leverage the predicted results for transient stability control. This article aims to bridge this gap and develop a new deep learning-based surrogate model for transient stability preventive control.

Preventive control aims to ensure the reliable operation of power systems in a future horizon, e.g., the next 15 min, by triggering appropriate actions in advance, such as generation redispatch, load shedding, and topology reconfiguration. The preventive control is derived from the optimal power flow (OPF) model with several constraints considering predefined contingencies. Transient stability-constrained OPF (TSC-OPF) is one of the most widely used approaches to determine preventive actions with consideration of stability margins in the presence of multiple faults [15]. However, TSC-OPF is computational-prohibitive for online preventive control. To deal with that, in [16], the trajectory sensitivity is leveraged on the one-machine infinite bus system. But the trajectory sensitivity could not handle multiobjective optimization problems. In [17], the dynamic secure/insecure regions of the system are determined through pattern recognition for preventive dynamic security control. However, the key features are manually extracted, which may yield degraded performance for more complicated scenarios. It is worth pointing out that very few machine learning models [18] have been used for preventive control in the literature and they are computational demanding for practical applications [19], or only for single-objective optimization without considering the cost and other power systems operational status [20]–[23]. This restricts their applicability of dealing with multiobjectives, such as balancing the minimization of generator dispatch costs and minimizing load shedding.

In this article, a new deep-learning-enabled computationally efficient transient stability preventive control framework is proposed. It has the following two main contributions.

1) The deep belief network (DBN) is strategically integrated with the reference-point-based nondominated sorting genetic algorithm (NSGA-III) to develop the new multiobjective optimization framework for preventive control. This allows us for the first time to connect the transient stability predictions from deep learning with the multiobjective optimizations, including minimizing the costs of generator redispatch and load shedding. Specifically, DBN maps the relationship between power system operational features and transient stability margin constrained by predefined contingencies. This leads to the development of a surrogate model for physical transient stability constraints. The latter are traditionally modeled by a large set of differential-algebraic equations and are the main computing bottleneck. The surrogate model significantly reduces the computational complexity of the full TSC-OPF by simply performing direct neural network-based transient stability verification in each iterative searching step for the optimal preventive control strategy, rather than executing many time-domain simulations.
2) The traditional approaches, such as the interior point method and the convex programming, can hardly address the optimization problem with implicit black-box constraints derived from neural networks [24]. To deal with black-box constraints, the NSGA-III algorithm that can find the Pareto-optimal front of multiobjective is integrated with the DBN enabled the surrogate model to coordinate the multiple objectives. Comparative results with other methods demonstrate that the new framework achieves significantly improved computational efficiency without loss of control performance.

The rest of this article is organized as follows. Section II presents the DBN-based surrogate modeling methodology. Section III shows the proposed framework. Numerical results are presented and analyzed in Section IV. Finally, Section V concludes this article.

II. Surrogate Modeling Methodology

A. Generation of Data Samples

Preventive control aims to change the system operating point by shifting generator outputs and/or load shedding to avoid the risk of violating security criteria. Note that the knowledge of generator voltage magnitudes and real power injections, as well as the real power injections and real and reactive power injections of all loads, is sufficient to determine the initial conditions of all generators for transient stability simulations. The latter determines the transient stability index (TSI). Therefore, the operational features $\mathbf{\Lambda} = (\mathbf{P_G}, \mathbf{V_G}, \mathbf{P_L}, \mathbf{Q_L})^T$ that include the voltage magnitudes of the generators $\mathbf{V_G}$, the active power of the generators $\mathbf{P_G}$ and loads $\mathbf{P_L}$, and reactive power of the loads $\mathbf{Q_L}$ are selected as the input features for DBN. To achieve high accuracy and good generalization, the data samples for training DBN should contain all possible operational features. This means that the data samples should be representative and evenly distributed in the sampling space. In this article, the Latin hypercube sampling (LHS) is applied to generate data samples associated with each operation scenario at the same time.

LHS is a multidimensionally distributed statistical method that generates random samples of the focused variables. Latin square (LS) refers to a square grid with one sample in each row and column, and Latin hypercube (LH) is the generalization of LS in the multidimensional scenario. Let N_S denote the number of samples, LHS divides each dimension of sample space into N_S equal parts and then gets sampling according to the LH principle. This sampling method ensures the independence and uniformity of data samples.

Besides varying generation-load balance in the sampling process, the permanent three-phase short-circuit faults on given lines are considered as pre-contingencies for each scenario. Through extensive off-line time-domain simulations, we can obtain generator rotor angle trajectories and transient stability index, TSI. TSI reflects the maximum rotor angle difference of the generators in the transient process and its mathematical expression is [17]

$$\mathrm{TSI} = \frac{360 - |\delta_{\max}|}{360 + |\delta_{\max}|} \times 100 \tag{1}$$

$$\delta_{\max} = \max_{t \in T}(\delta_{i,t} - \delta_{j,t}) \tag{2}$$

where $\delta_{\max}$ is the maximum rotor angle difference between any two generators; and T is the observation interval. When TSI $>$ 0, the system is transient stable while TSI < 0 means system unstable. Note that the increase of TSI reflects transient stability enhancement and vice versa.

Assuming that the number of samples for the operational features generated by the LHS is k and the number of contingencies is p, then $k{*}p$ time-domain simulations are performed. To ensure that DBN can predict the TSI corresponding to the most severe scenario derived from the predefined contingencies, the operational features and the minimum TSI of p TSIs are selected to form the sample sets for DBN training. It should be clarified that the scope of the article is a preventive control. It is well known that preventive control aims to deal with severe contingencies with high probability and thus could not deal with all possible contingencies. If preventive controls are not effective in addressing some unexpected events, emergent controls are needed, which is out of the scope of this article.

B. Deep Belief Network-Based Transient Stability Predictor

DBN is a probability generation model, which is composed of a multilayer restricted Boltzmann machine (RBM) and a fully connected layer. RBM is a two-layer unsupervised network that contains a visible layer and a hidden layer. The main feature of RBM is its ability to probabilistically reconstruct input features by two-way connections.

DBN learning has two main steps: pretraining and fine-tuning. Pretraining solves the problems of gradient disappearance and the explosion of the deep neural network. In the pretraining step, RBM is trained layer by layer through the input data. Then, DBN reconstructs multiple features of input data. In the fine-tuning step, DBN further trains the model with supervision to fit input data and labels.

RBM is an energy-based model. Defining *v* and *h* the visible layer and the hidden layer matrices, respectively, the energy of a configuration (*v, h*) is

$$E(v,h) = -a^T v - b^T h - v^T W h \tag{3}$$

where *a* and *b* are the bias matrices of the visible layer and the hidden layer; and *W* is the weight matrix between the visible layer and the hidden layer. The probability distributions of *v* and *h* can be obtained from the energy of (*v, h*)

$$P(v,h) = \frac{1}{Z} e^{-E(v,h)} \tag{4}$$

$$Z = \sum_v \sum_h e^{-E(v,h)} \tag{5}$$

where *Z* is the partition function, also known as the normalizing constant, the aim of which is to ensure the sum of the probability distribution is 1.

From (3), the marginal probability of the visible layer vector can be calculated by accumulating all possible hidden layer configurations via

$$P(v) = \frac{1}{Z} \sum_h e^{-E(v,h)}. \tag{6}$$

Since RBM lacks intralayer connections, the activations of the visible units and hidden units are independent. In other words, given the configuration of the hidden unit *h*, the conditional probability of the visible unit *v* is

$$P(v|h) = \prod_{i=1}^{m} P(v_i|h). \tag{7}$$

Similarly, the conditional probability of *h* given *v* is

$$P(h|v) = \prod_{j=1}^{n} P(h_j|v). \tag{8}$$

The individual activation probabilities are given by

$$P(h_j = 1|v) = \mathrm{SeLU}\left(\sum_{i=1}^{m} w_{i,j} v_i + b_j\right) \tag{9}$$

$$P(v_i = 1|h) = \mathrm{SeLU}\left(\sum_{j=1}^{n} w_{i,j} h_j + a_i\right) \tag{10}$$

$$\mathrm{SeLU} = \begin{cases} x, & x \geq 0 \\ \alpha(e^x - 1), & x < 0 \end{cases} \tag{11}$$

where SeLU is the activation function and α is the scale factor.

Training RBM by maximizing the expected log probability of a few training sets *V* yields

$$\arg\max_W \prod_{v \in V} P(v). \tag{12}$$

Taking *w* as an example, the partial derivation of *w* can be obtained as follows:

$$\frac{\partial \ln(P(v))}{\partial w_{i,j}} = P(h_j = 1|v_t) v_i^t - \sum_v P(v) P(h_j = 1|v) v_i. \tag{13}$$

The weight $w_{i,j}$ is updated by

$$w_{i,j} = w_{i,j} + \eta \frac{\partial \ln(P(v))}{\partial w_{i,j}} \tag{14}$$

where η denotes the learning rate.

Similarly, the energy function and probability distribution function of DBN are as follows:

$$E(v,h^1,h^2,h^3) = -v^T W^1 h^1 - {h^1}^T W^2 h^2 - {h^2}^T W^3 h^3 \tag{15}$$

$$P(v,h^1,h^2,h^3) = \frac{1}{Z} e^{-E(v,h^1,h^2,h^3)}. \tag{16}$$

In this article, the TensorFlow framework is used to build the DBN model. The operational features are taken as input vectors, and the TSIs are used as the corresponding labels to learn the nonlinear mapping relationship between them. After training, TSI can be obtained quickly through transient stability predictor, which can be expressed as follows:

$$\mathrm{TSI}_P = \mathcal{L}(\boldsymbol{\Lambda}) \tag{17}$$

where $\mathcal{L}(\boldsymbol{\Lambda})$ is the transient stability predictor and its input includes the operational features while its output is the TSI prediction, called TSI_P.

When training the DBN-based predictor, normalization is used to regularize data of different dimensions. The learning rate decay algorithm is used to speed up the learning speed and accuracy. In addition, L_2 regularization is used to prevent model overfitting. Their mathematical expressions are

$$F_{i,N} = \frac{F_i - F_{i,\min}}{F_{i,\max} - F_{i,\min}} \tag{18}$$

$$\eta = \eta_0 \times \frac{1}{1 + D \times E} \tag{19}$$

$$LF = \frac{1}{2r} \sum_{\boldsymbol{\Lambda} T} \left\| \mathcal{L}(\boldsymbol{\Lambda}_T) - \mathrm{TSI}_T \right\|^2 + \frac{\lambda}{2r} \sum_w w^2 \tag{20}$$

where F_i and $F_{i,N}$ denote, respectively, the original and normalized data of the *i*th dimension data; $F_{i,\max}$ and $F_{i,\min}$ are the maximal and minimal values of F_i; η is the learning rate; η_0 is the initial value of η; *D* is the learning rate attenuation coefficient; *E* is epoch numbers; *LF* is the loss function with L_2 regularization, in which the first part is the mean square error (MSE) and the second part is the L2 regularization loss; *r* is the size of the training set; $\mathbf{\Lambda}_T$ is the operational features of the test set; TSI_T is the TSI of the test set; and λ is a regularization parameter.

In this article, the accuracy of the predictor is calculated by the following equation:

$$\begin{cases} A_{\mathrm{PC}} = 1 - \frac{|(\mathrm{TSI}_T<0)\wedge(\mathrm{TSI}_P>\mathrm{TSI}_B)|}{N_T} \\ A_{\mathrm{DBN}} = \frac{|\mathrm{SIGN}(\mathrm{TSI}_T)\leftrightarrow\mathrm{SIGN}(\mathrm{TSI}_P)|}{N_T} \end{cases} \tag{21}$$

where A_{PC} is the prediction accuracy of the preventive control; $\wedge$ is the logical conjunction; N_T is the size of the test set; note that the second part after the negative sign represents the proportion that the scenarios predicted to be stable but should be unstable; A_{DBN} is the prediction accuracy of DBN; $\leftrightarrow$ is the logical biconditional; SIGN is a function to find the sign of a value; and TSI_B is the critical TSI to assess the transient stability of power system.

C. Surrogate-Assisted TSC-OPF

Traditional preventive control relies on the TSC-OPF model with predefined contingency constraints. The aim is to take appropriate actions to change the power system operation point while achieving the optimal objective subject to all constraints. However, the transient stability constraints consist of a set of differential-algebraic equations, which are computationally extensive to assess for online applications. To address that, the DBN-based predictor, for the first time, is introduced to surrogate the most computationally complicated parts of the TSC-OPF model. The surrogate model is much more computationally efficient than the original time-domain simulations. Specifically, the surrogate-assisted preventive control model is presented as follows.

1) Objective Function: It aims to minimize the cost and load ratio of system components, especially generators and transmission lines.

$$F_{\mathrm{cost},t} = \sum_{i\in S_g} \left(C_{Ui}\left|\Delta P_{Ui,t}\right| + C_{Di}\left|\Delta P_{Di,t}\right|\right) + \sum_{j\in S_{ls}} \left(C_{Lj}\left|\Delta P_{Lj,t}\right|\right) \tag{22}$$

$$R_{a,t} = \frac{1}{M}\sum_{k=1}^{M} \frac{|P_{RLk,t}|}{P_{RCk,t}} \times 100\% \tag{23}$$

$$\Delta P_{Ui,t} = \begin{cases} P_{Gi,t} - P_{Gi,t-1}, & P_{Gi,t} > P_{Gi,t-1} \\ 0, & P_{Gi,t} \le P_{Gi,t-1} \end{cases} \tag{24}$$

$$\Delta P_{Di,t} = \begin{cases} 0, & P_{Gi,t} \ge P_{Gi,t-1} \\ P_{Gi,t} - P_{Gi,t-1}, & P_{Gi,t} < P_{Gi,t-1} \end{cases} \tag{25}$$

$$\Delta P_{Lj,t} = P_{Lj,t-1} - P_{Lj,t} \tag{26}$$

where *t* represents time instant; *i* is the generator number; *j* is the load number; *k* is the component number; *M* is the number of components; $F_{\mathrm{cost},\,t}$ is the total cost; C_{Ui} and C_{Di} are the costs of increasing and decreasing the generator outputs; C_{Lj} is the cost of load shedding; $\Delta P_{Ui,t}$ and $\Delta P_{Di,t}$ are the sizes of increasing and decreasing the generator outputs; $\Delta P_{Lj,t}$ is the size of the load shedding; $P_{RLk,t}$ is the current load of component *i*, $P_{RCk,t}$ is the capacity of component *i*; $R_{a,t}$ is the average load ratio of components, including the average load ratio of generators and transmission lines; S_{ls} denotes the set of loads that can be shed; and S_g denotes the generator set.

2) Equality Constraints: Power flow equations:

$$\begin{aligned} P_{Gi,t} &= P_{Li,t} + \sum_{j=1}^{n} |V_{i,t}|\,|V_{j,t}|\left(G_{ij}\cos\alpha_{ij,t} + B_{ij}\sin\alpha_{ij,t}\right) \\ Q_{Gi,t} &= Q_{Li,t} + \sum_{j=1}^{n} |V_{i,t}|\,|V_{j,t}|\left(G_{ij}\sin\alpha_{ij,t} - B_{ij}\cos\alpha_{ij,t}\right) \\ & \qquad i,j \in S_{\mathrm{n}} \end{aligned} \tag{27}$$

where P_{Gi} and Q_{Gi} are active and reactive power generations at bus *i*; P_{Li} and Q_{Li} are active and reactive load injections at bus *i*; $|V_i|$and $|V_j|$ are voltage magnitudes at bus *i* and *j*; α_{ij} is bus voltage angle difference; G_{ij} and B_{ij} are the real and imaginary parts of the line admittance; and S_n denotes the bus set.

3) Inequality Constraints: Operational constraints:

$$\begin{cases} P_{Gi}^{\min} \le P_{Gi,t} \le P_{Gi}^{\max}, i \in S_g \\ Q_{Gi}^{\min} \le Q_{Gi,t} \le Q_{Gi}^{\max}, i \in S_g \\ V_i^{\min} \le V_{i,t} \le V_i^{\max}, i \in S_n \\ I_{ij}^{\min} \le I_{ij,t} \le I_{ij}^{\max}, (i,j) \in S_l \\ -R_{Di} < (P_{Gi,t} - P_{Gi,t-1})/\Delta t < R_{Ui}, i \in S_g \end{cases} \tag{28}$$

where $P_{Gi}^{\max}$ and $P_{Gi}^{\min}$ are the upper and lower limits of generation; $Q_{Gi}^{\max}$ and $Q_{Gi}^{\min}$ are the upper and lower limits of the reactive power outputs; $V_i^{\max}$ and $V_i^{\min}$ are the upper and lower limits of bus voltages; $I_{ij}^{\max}$ and $I_{ij}^{\min}$ are the boundaries of thermal limits of line *ij*; S_l denotes the line set; R_U and R_D are ramp ratios corresponding to increasing and decreasing generator outputs; and Δt is preventive control interval.

4) Transient Stability Constraint: Traditional transient stability constraints are $|\delta_{i,t} - \delta_{i,t}^{COI}| \le \delta^{\max}, i \in S_c$ obtained from time-domain simulations and are computationally prohibitive. To deal with that, it is surrogated by the following proposed computationally efficient transient predictor:

$$\mathcal{L}(\mathbf{\Lambda}_t) > \mathrm{TSI}_B \tag{29}$$

where $\mathcal{L}(\cdot)$ is the DBN-based surrogate operator. When $\mathrm{TSI} > \mathrm{TSI}_B$, the system is transient stable for the given operational features and vice versa. $\mathcal{L}(\cdot)$ provides the fast-enough transient stability awareness in every searching iteration close to the optimal preventive control actions. The proposed DBN-based surrogate model for TSC-OPF is shown in Fig. 1.

III. Solution Methodology

The traditional approaches, such as the interior point method and the convex programming, could not address the formulation with implicit black-box constraints [24], such as the surrogate model in this article. Meanwhile, the widely used evolutionary

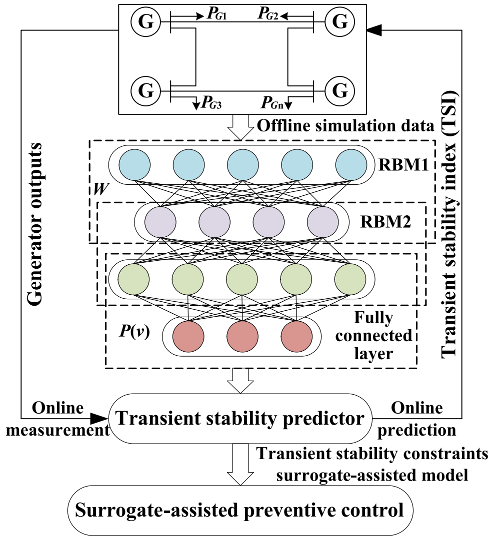

Fig. 1. DBN based surrogate model for multiobjective TSC-OPF.

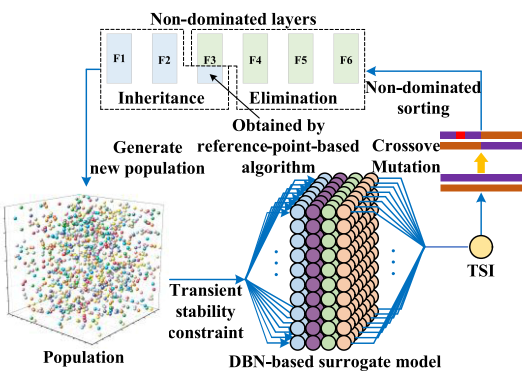

Fig. 2. Proposed surrogate model-assisted preventive control using NSGA-III algorithm.

TABLE I
IMPLEMENTATION OF NSGA-III ALGORITHM

Algorithm 1 NSGA-III ALGORITHM FOR OPTIMIZATION

Input: Transient unstable scenarios.

Initialize: Set the fluctuation ranges of the generator outputs and loads in the NSGA-III algorithm and randomly generate initial parent population NP_1 of generator outputs and loads with size N that are within these ranges.

Repeat:

1 At the ith iteration, generate the offspring population NQ_i with size N by NP_i through crossover and mutation, where NP_i and NQ_i are combined to generate a population K_i with a size of $2N$.

2 Based on the power flow calculations and DBN-based surrogate model, the multi-objective values of solutions in K_i are obtained, where (22)-(26) and (30)-(31) are used.

3 Use non-dominated sorting to divide K_i into several non-dominated layers (F_1, F_2, etc.).

4 In sequence of the non-dominated layer, the solutions of each non-dominated layer are added to a new population NP_{i+1} until the size of NP_{i+1} is equal to N, or greater than N for the first time when adding solutions in non-dominated layer F_l. The solutions in the F_{l+1} and later layers will be eliminated.

5 The insufficient solutions of NP_{i+1} are selected from F_l through the following steps: ***a.*** Determining the reference points on the hyperplane; ***b.*** Adaptive normalization of population solutions; ***c.*** Associating each population solution with a reference point; ***d.*** Niche-preservation operation and ***e.*** fill all solutions of NP_{i+1}.

Stop: The preset number of iterations is reached.

Output: Optimal preventive control strategies (F_1 of the last iteration).

algorithms, such as bat algorithm, firefly algorithm, and clonal selection algorithm, only handle single objective optimization and therefore not eligible for the multiobjective optimization in this article. To deal with that, the NSGA-III [25] is adopted, the advantages of which as compared to other multiobjective optimization algorithms have been proved. NSGA-III can find the multiobjective Pareto-optimal front. The proposed solution methodology for the surrogate-assisted preventive control is shown in Fig. 2. The optimization model aims to minimize the following four objectives simultaneously subject to equality and inequality constraints (27) and (28).

1) The total cost, including the costs of shifting the generator outputs and load shedding according to preventive control strategy; it is calculated using (22)–(26).
2) The index TSI

$$\mathrm{TSI}_P = \begin{cases} \mathrm{TSI}_B & ,\mathcal{L}(\mathbf{\Lambda}^{pre}) > \mathrm{TSI}_B \\ \mathcal{L}(\mathbf{\Lambda}^{pre}) & ,\mathcal{L}(\mathbf{\Lambda}^{pre}) < \mathrm{TSI}_B \end{cases} \tag{30}$$

$$\mathbf{\Lambda}^{pre} = (\mathbf{P}_{\mathbf{G}}^{pre}, \mathbf{V}_{\mathbf{G}}, \mathbf{P}_{\mathbf{LS}}^{pre}, \mathbf{P}_{\mathbf{LR}}, \mathbf{Q}_{\mathbf{L}})^T \tag{31}$$

where $\mathbf{P}_{\mathbf{G}}^{pre}$ represents the generator outputs given by the preventive control strategy; $\mathbf{P}_{\mathbf{LS}}^{pre}$ denotes the active power of loads that can be shed; $\mathbf{P}_{\mathbf{LR}}$ is the active power of remaining loads; except for $\mathbf{P}_{\mathbf{G}}^{pre}$ and $\mathbf{P}_{\mathbf{LS}}^{pre}$, the other operational features are initial values, which are used together to form $\mathbf{\Lambda}^{pre}$; TSI_P larger than TSI_B is set to TSI_B instead of pursuing higher TSI_P to reduce control cost.

1) The third and fourth optimization objectives are to reduce the load ratio of generators and transmission lines. The average load ratio can be calculated using (23).

NSGA-III uses a set of well-spread reference points instead of crowding degree ranking to maintain population diversity as compared to the NSGA-II algorithm [25]. This leads to improved performance. At each iteration, the worst-case computational complexity of NSGA-III is the larger one of $O(N^2\log^{M-2}N)$ or $O(N^2M)$, where M is the dimension of the objectives. Its implementations can be found in Table I.

In the NSGA-III algorithm, the TSI of the population is repeatedly calculated using the DBN-based transient stability predictor (the surrogate model) during the optimal preventive control strategy searching process. Assuming DBN has u layers, only u-1 matrix multiplications and u-2 activation function calculations are needed to predict TSI. Thus, the surrogated-assisted model allows us to calculate TSI in a very efficient manner. In the meanwhile, DBN can input all solutions of a population in the form of a matrix and this enables us to get TSIs of all solutions through the same calculation process with a single TSI calculation. This greatly improves the convergence speed of the NSGA-III algorithm.

It is worth pointing out that only the optimal solution that meets the actual stability demands of the power system will be selected and the selection criteria are as follows.

1) Eliminate solutions that violate any individual constraint.
2) Transient stability is the primary objective of preventive control. Therefore, the solutions are selected according to the following aspects: (a) in most studies, $\text{TSI} > 0$ generally means the system is transient stable, but in fact, when TSI is slightly higher than 0, the system is located at critical stability point. With the increase of simulation time, the power system is likely to change from the critical stability to instability. Thus, based on the conservatism principle of preventive control, a constant TSI_B that is greater than 0 is selected, (b) in control theory, there are definitions of stable, critical stable, and unstable. The definition of critical unstable is based on conservative control. For example, in [18], the boundary point between stable and critical stable is defined as $\text{TSI} = 10$. In this article, a sensitivity test, which investigates the relationship between TSI_B and A_{PC}, is performed. According to the results, TSI_B is selected as 60 to make A_{PC} reach 100%. At this time, $\text{TSI} > \text{TSI}_B = 60$ means that when the predicted $\text{TSI} > 60$, the system is considered transient stable, while the predicted $\text{TSI} < 60$ is considered unstable and preventive control should be performed to stabilize the system. Note that that in a special situation, such as heavy loading, the actual demand can be met by reducing the stability criterion, i.e., $\text{TSI} > \text{TSI}_B = 30$ can be considered; and (c) a too large TSI increase means a large cost. To this end, the dispatcher will comprehensively consider the TSI, the total cost, the load ratio of the generators, and transmission lines when determining the optimal preventive control strategy.

The detailed steps for the DBN-based surrogate model for preventive control are summarized in Fig. 3.

IV. Case Study

In this section, the test system and data samples for training will be first shown. Then, the validity of the DBN for surrogate modeling will be demonstrated via simulations and comparison results with other alternatives. Finally, the integration of DBN results with the optimal model for generator redispatch and load shedding decision-making is analyzed.

A. Test Systems and Generation of Data Samples

The proposed method is tested on IEEE 39, 68, and 140-bus systems. First, data samples need to be generated. For three test systems, generator outputs and active power of the loads are assumed to fluctuate ranging from 80 to 120%; the voltage magnitudes of the generators are assumed to fluctuate in the range of 1.0–1.05, and the reactive power of the loads is derived from the active power to ensure the constant power factor. Considering the severity of the contingencies according to historical operational experience, 13, 19, and 31 three-phase short-circuit faults with fault duration of 0.1s are selected as contingencies for three test systems. Then, for three test systems, the sample sizes are set to be 2000, 3000, and 4000, respectively, according to the sensitivity analysis shown in Fig. 4, and the LHS is used to generate the corresponding operational scenarios. For each test system, operational features and contingencies are combined to form 26 000, 57 000, and 124 000 transient fault scenarios, respectively. The time-domain simulation period is 20 s. For each scenario, the vector of operational features combined with its corresponding minimum TSI is treated as one data sample. Note that for systems with larger scales and more complex structures, more samples are needed to cover diverse scenarios depending on the number of credible contingencies. This justifies the choice of sample size and this number can be increased if the system complexity increases and when more credible contingencies need to be considered.

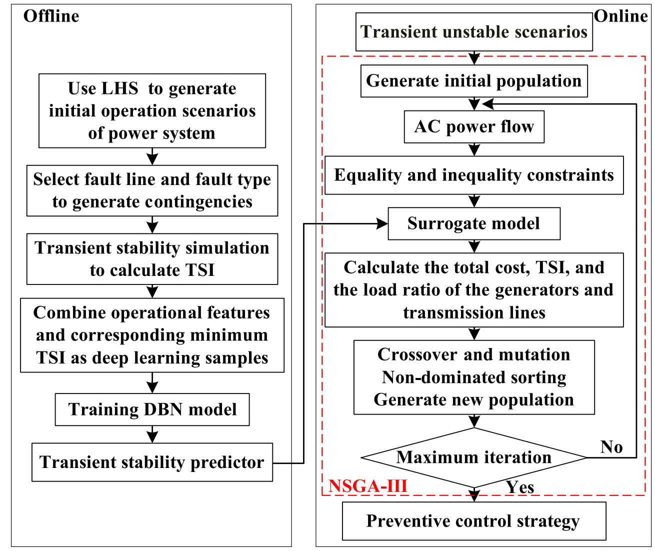

Fig. 3. Proposed DBN-based surrogate model for preventive control.

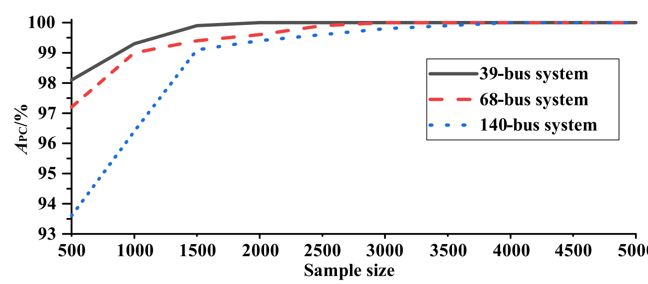

Fig. 4. Sensitivity analysis of the relationship between sample size and A_{PC}.

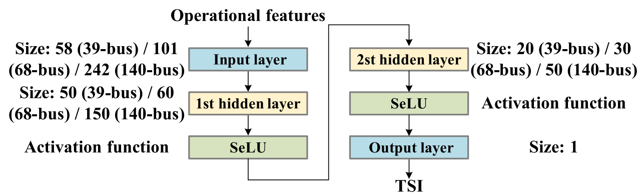

Fig. 5. Structure of the proposed DBN-based transient stability predictor.

TABLE II
ACCURACY OF TRANSIENT STABILITY PREDICTOR

Test system	Prediction accuracy A_{PC}		
	$\mathrm{TSI}_B = 0$	$\mathrm{TSI}_B = 60$	A_{DBN}
39-bus system	99.4%	100%	96.3%
68-bus system	98.9%	100%	93.8%
140-bus system	99.0%	100%	93.9%

B. Comparisons of Transient Stability Predictors

The samples are randomly divided into a training set and a test set, accounting for 80% and 20%, respectively. The training set is used to train the transient stability predictor while the test set is for validation. Then, (18) is used for data sample normalizations. The three predictors corresponding to three test systems have four layers with a number of neurons 58-50-20-1, 101-60-30-1, and 242-150-50-1, respectively, and batch size 5, epoch 100, initial learning rate 0.01, and activation function SeLU. The structure of the predictor is shown in Fig. 5. The prediction accuracies (A_{PC} and A_{DBN}) for different test systems are shown in Table II. For A_{PC}, two TSI_B cases are considered and tested. It can be found that the proposed predictor achieves the best performance of preventive control for different systems, where the results for $\mathrm{TSI}_B = 60$ are more conservative than that for $\mathrm{TSI}_B = 0$. This demonstrates the effectiveness of the proposed predictor for surrogate modeling. Also, TSI_B can be set according to actual needs by modifying (30) in the NSGA-III algorithm.

The ANN, SVM, convolutional neural network (CNN), and recurrent neural network (RNN) are also constructed to compare their performances with the proposed predictor. The first ANN consists of three layers, and the number of neurons is 10-50-1, i.e., ANN^3. The second ANN consists of four layers, which adopt the same network structure and has the number of neurons the same as the predictor, i.e., ANN^4. The optimizer, activation function, loss function, and data preprocessing method are the same as the proposed predictor. The SVM is constructed by scikit-learn, which is a machine learning package in Python. The CNN and RNN are constructed by Tensorflow. Specifically, one-dimensional CNN (1D CNN) is used, where the convolution window is 7, max pooling is used, and activation function is set to ReLU. RNN selects long short-term memory (LSTM) with dropout and ReLU. Note that both 1D CNN and LSTM have the same numbers of layers, iterations and optimizers as DBN. The prediction accuracies and losses for six methods in three test systems are shown in Fig. 6, where the loss is characterized by the MSE of TSI_T and TSI_P after normalization. From Fig. 6, we can observe that DBN has the fastest convergence speed, minimal test set loss and the best prediction performance as compared to the other five alternatives. This is mainly due to the pretraining step that allows DBN to achieve multiple feature reconstructions from the input data. It also solves the problems of gradient disappearance and gradient explosion of deep neural networks, such as ANN^4. It is worth noting that CNN is mainly used in image recognition, focusing on the local information of the image, while RNN is widely used in natural language processing. It mainly focuses on sequence data, such as time-series data. They are popular methods in their respective fields, but not suitable for the application here according to the test results.

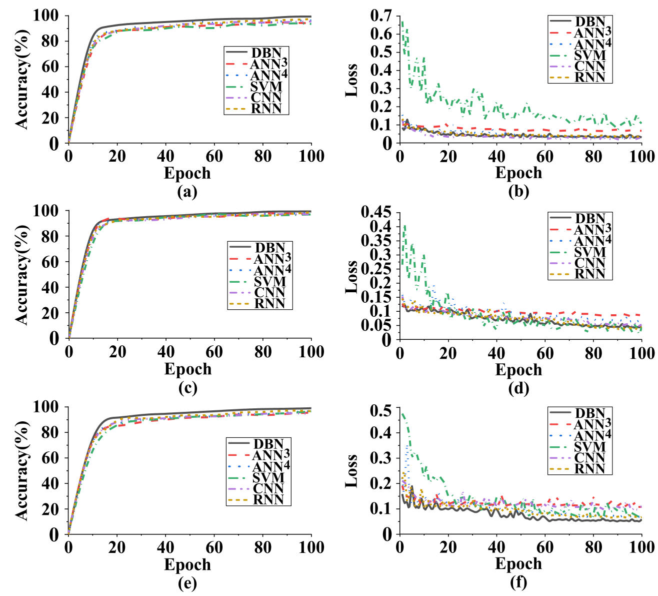

Fig. 6. Comparison results for DBN, ANN^3, ANN^4, SVM, CNN, and RNN. (a) and (b) Prediction accuracy and loss in the 39-bus system. (c) and (d) Prediction accuracy and loss in the 68-bus system. (e) and (f) Prediction accuracy and loss in the 140-bus system.

C. Preventive Control Without Load Shedding

As shown in Fig. 2, the DBN-based surrogate model is embedded in the iterative optimization process of the NSGA-III algorithm to search preventive control to avoid instability considering predefined contingencies. In order to determine the parameters of the NSGA-III algorithm, the population sizes are set to 100, 200, and 300, called S_{100}, S_{200}, and S_{300} respectively, and their comparison results are shown in Table III. From Table III, we can see that S_{200} and S_{300} are the nondominated solutions as compared to that of S_{100}. Therefore, the population size needs to be larger than 100. It should be noted that the differences between the solutions of S_{200} and S_{300} are minor while the one with a smaller population size has a shorter iteration time. Therefore, the population size is set to 200. To further investigate the impacts of setting different numbers of iterations of NSGA-III,

TABLE III
CHOICE OF NONDOMINATED DIFFERENT POPULATION SIZES

Objectives	Population size 100	200	300
TSI	60	60	60
Total cost	1818	1146	1773
Generators load ratio	62.5	62.4	61.6
Transmission lines load ratio	44.9	44.4	44.3

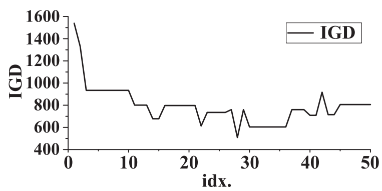

Fig. 7. Inverse generational distance at a population size of 200.

TABLE IV
COMPARISON OF ONLINE COMPUTATIONAL EFFICIENCY

Algorithm	Computation time 39-bus system	68-bus system	140-bus system
NSGA-III with DBN surrogate model	47s	49s	58s
NSGA-III with time-domain simulation	>40min	>1.5h	>2.5h

the inverse generational distance (IGD) curve that can evaluate the convergence and distribution performance is used and its corresponding results with different iteration numbers are shown in Fig. 7. Since a good solution set corresponds to a small IGD, thus the number of iterations is set as 30 according to the results shown in Fig. 7. To make the reference points widely distributed on the entirely normalized hyperplane, the number of reference points is set to 10 000. The online computational efficiencies of the NSGA-III algorithm for the surrogate model-based and the original time-domain simulation-based methods for three systems are compared and the results are displayed in Table IV. It can be found that the use of the DBN-based surrogate model can significantly improve the computational efficiency of the NSGA-III algorithm, i.e., more than 30 times faster. It is also interesting to find that, unlike the physical-model-based TSC-OPF, the computing time of the proposed preventive control method only slightly increases with the increased scale of power systems. This is because system size has a negligible effect on some operations of the NSGA-III algorithm, such as non-dominated sorting that accounts for a large percentage of the total calculation time. The offline process includes time-domain simulations to generate training samples and surrogate model training (185.6, 189.2, and 198.7 s, respectively). The time of the online application is shown in Table IV. It should be noted that the offline process only needs to be performed once, unless the system has significant changes, such as topology changes beyond $N-1$. Once the training is done, the proposed method only needs less than 1 min for preventive control.

Furthermore, the generator dispatch results obtained from the preventive control strategy for the three test systems are shown in Fig. 8. It can be observed that the generator outputs slightly fluctuate before preventive control. This is because the NSGA-III algorithm can control the generation dispatch cost and reduce the adjustment range of generator outputs while maintaining system stability constraints. Setting C_U and C_D, according to (22)–(26), the total costs of 39, 68, and 140-bus systems are \$1146, \$4912, and \$6582, respectively. The detailed calculation processes of the total cost of the 39-bus system are shown in Table V. It can be observed that Gen. 10 has the maximum power output increase, i.e., 65.6 MW, while it has a lower C_U, indicating that preventive control tends to regulate the generator with a smaller cost. Similarly, Gen. 2 with lower C_D has a larger output decrease. After that, the transient stability for each contingency can be guaranteed; see Table VI for more details.

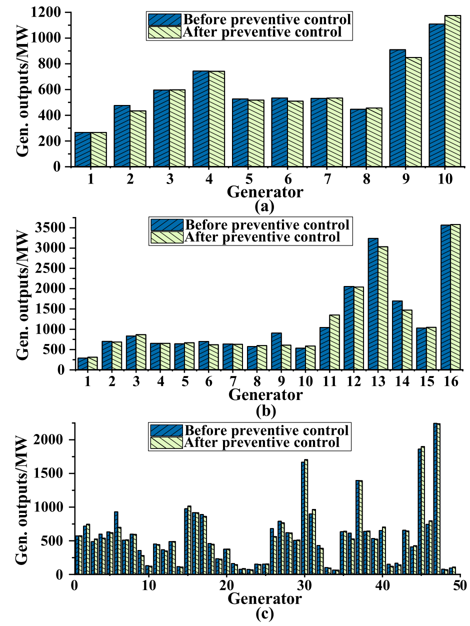

Fig. 8. Generator redispatch results without load shedding. (a) 39-Bus system. (b) 68-Bus system. (c) 140-Bus system.

TABLE V
TOTAL COSTS OF 39-BUS SYSTEM

Gen.	Before control /MW	After control /MW	Output change /MW	Unit up cost/\$	Unit down cost/\$	Gen. cost/\$	Total cost/\$
1	266.7	266.5	-0.2	8.78	3.42	0.68	1146
2	**476.4**	**434.2**	**-42.2**	**8.81**	**2.72**	**114.78**	
3	596.9	597.7	0.8	9.81	7.54	7.85	
4	743.9	742.1	-1.8	8.42	4.97	8.95	
5	527.1	518.2	-8.9	10.75	3.53	31.42	
6	534.9	509.8	-25.1	13.40	3.27	82.08	
7	531.1	534	2.9	8.76	8.16	25.4	
8	446.4	456.5	10.1	10.17	8.21	102.72	
9	909.8	848.8	-61	7.96	4.20	256.2	
10	**1109.9**	**1175.5**	**65.6**	**7.87**	**3.36**	**516.27**	

The significance of bold entities is to highlight numbers, which have the highest Gen. cost/\$.

To validate the effectiveness of the proposed controls, the time-domain simulation results are displayed. The preventive

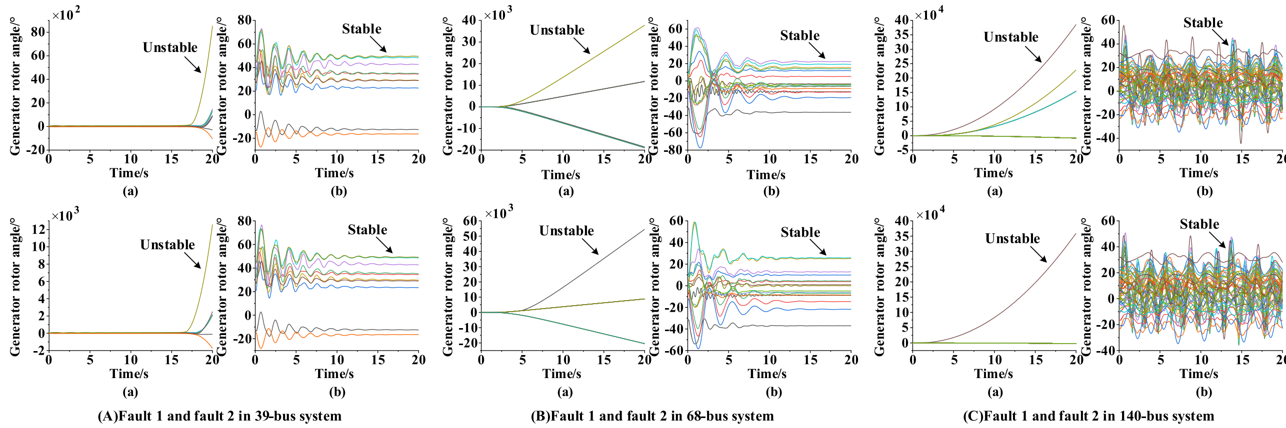

Fig. 9. Comparison of rotor angle trajectories before and after preventive control in three test systems.

TABLE VI
COMPARISONS OF TSI BEFORE AND AFTER PREVENTIVE CONTROL

Fault ID	39-bus system Before control	After control	68-bus system Before control	After control	140-bus system Before control	After control
1	-92.8	69.1	-98.7	72.0	-99.8	75.0
2	-95.1	69.2	-99.0	70.3	-99.8	75.2
3	-97.5	70.1	-98.9	78.5	-99.8	74.8
4	-11.9	69.5	-96.5	74.4	-99.8	70.4
5	-87.5	68.7	-96.1	72.2	-99.8	62.0
6					-99.7	70.0

control results for those faults that cause system instability are displayed in Table VI.

From Table VI, we can see that after performing the proposed preventive control, TSIs for all the contingencies are greater than TSI_B. For example, in the 39-bus system, the unstable cases, i.e., fault 1 (TSI $= -92.8$) and fault 2 (TSI $= -95.1$), are controlled to be stable with the value of index higher than 60. To achieve this, the traditional approach needs nearly an hour, while the proposed method only spends less than 1 min (see Table IV). The comparisons of rotor angle trajectories corresponding to the first two fault scenarios in Table VI before and after preventive control in three test systems are shown in Fig. 9. It is evident that all diverging rotor angle trajectories before preventive control have now been turned into stable cases, demonstrating the effectiveness of the proposed model. Furthermore, the load ratios of generators and transmission lines before and after preventive control are calculated using (23) and the results are shown in Table VII. It can be concluded that after preventive control, the load ratios of generators and transmission lines have been decreased significantly. Note that for each specific power networks, dispatchers can control the frequently overloaded devices to reduce the load ratio and ensure the safe operation of critical devices.

Finally, the Pareto-optimal front of TSI, total cost, and transmission line load ratio for the 39-bus system is shown in Fig. 10, where each point represents a solution of NSGA-III. The value of the point on each objective is mapped on the face of the

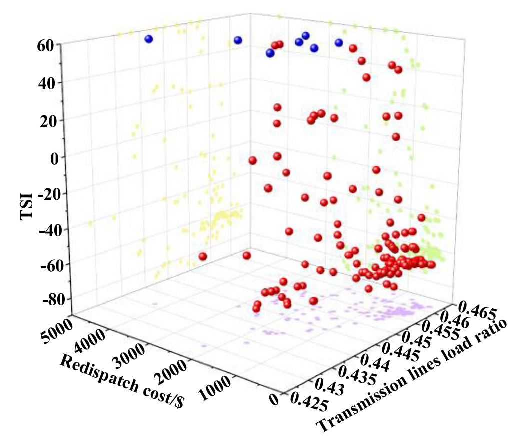

Fig. 10. Pareto-optimal front of the proposed method in the 39-bus system.

cube. It can be found that the TSIs of red points are less than 60, and therefore they will be discarded. All blue points are transient stability solutions. Note that a smaller transmission line load ratio corresponds to higher cost and dispatchers need to choose the best solution in the blue points according to the actual situation. When all constraints are met, the preventive control strategy with the lowest cost will be determined.

D. Preventive Control With Load Shedding

In some scenarios, such as heavy loading, generator redispatch may not be sufficient to guarantee system stability and load shedding is needed in the preventive control. Assuming that the three systems have 5, 6, and 10 loads that can be shed. Using the same scenarios as the previous case studies, the generator outputs and the loads which can be shed are used as the regulating variables in the NSGA-III algorithm, and the preventive control strategy is obtained. It is worth noting that the active power of the loads is always the input of DBN. When the preventive control without load shedding is performed, P_{LS}^{pre} represents the initial parameters. When the preventive control with load shedding is used, P_{LS}^{pre} is obtained by the preventive control strategy, namely

TABLE VII
COMPARISON OF LOAD RATIO BEFORE AND AFTER PREVENTIVE CONTROL

Load ratio (%)	39-bus system		68-bus system		140-bus system	
	Before control	After control	Before control	After control	Before control	After control
Generators	62.7	62.4	60.9	60.5	57.0	55.9
Transmission lines active power	46.4	44.4	45.7	42.0	40.1	39.8

TABLE VIII
COMPARISON OF OPTIMIZATION OBJECTIVES BEFORE AND AFTER PREVENTIVE CONTROL CONSIDERING LOAD SHEDDING

Optimization objectives	39-bus system		68-bus system		140-bus system	
	Before control	After control	Before control	After control	Before control	After control
Minimum TSI	-97.5	66.8	-99.0	65.2	-99.8	56.5
Load ratio of generators / %	62.7	61.6	60.9	52.8	57.0	56.3
Load ratio of transmission lines / %	46.4	44.1	45.7	32.3	40.1	38.6
Cost / $	2327		14887		15325	

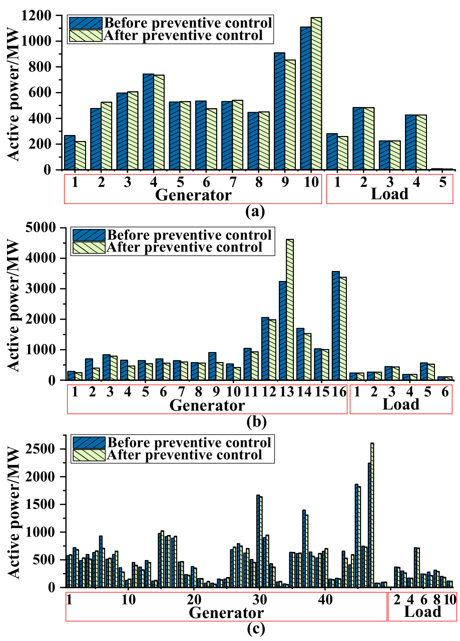

Fig. 11. Generator redispatch results considering load shedding. (a) 39-Bus system. (b) 68-Bus system. (c) 140-Bus system.

$P_{LS}^{pre} + \mathrm{P_{LR}} = \mathrm{P_L}$. As a result, there is no dimensionality issue as compared to the previous case shown in Section IV-C. It is worth noting that this is also an advantage of the proposed method since the DBN model does not need to be retrained regardless of the location and quantity of load shedding.

The corresponding results are shown in Fig. 11, where the comparisons of TSI before and after preventive control are displayed in Table VIII. As shown in Table VIII, by generator dispatch and load shedding, a minimum TSI similar to that of using generator dispatch is obtained (see results in Tables VI and VII). In terms of load ratio, due to load shedding, it is significantly better than generator dispatch only. However, in terms of cost, the cost per MW of load shedding is about two times of increasing the generator outputs and five times of decreasing the generator outputs. Thus, the cost is significantly higher than that of only generator dispatch. The results of preventive control considering load shedding show that the proposed method can effectively deal with different controlled variables and objectives.

E. Impacts and Countermeasures for DBN Mislabeling

When the proposed method is applied online, it is necessary to use DBN-based transient stability predictor to judge the risk of transient instability and decide whether to carry out transient stability preventive control. There are two cases of DBN mislabeling: classifying unstable scenario as stable one and classifying stable scenario as unstable one. For the first case, we can set a higher TSI_B to get conservative results depending on practical needs. As long as TSI_P is less than TSI_B, we can carry out preventive control to reduce the security risk with the increased cost. For the second case, an unnecessary preventive control will be implemented, yielding increased cost, which is the sum of costs for generation redispatch and load shedding. This has already been considered in the objective function (22) and the results are shown in Section IV-D. Therefore, in practical applications, TSI_B can be used to achieve the tradeoff between system safety and control cost.

F. Preventive Control Considering Topology Changes

Contingency analysis is typically studied for those credible ones that are well defined by the operators according to historical experiences. The focus of preventative control is usually used to stabilize the system if these contingencies lead to stability issues. Indeed, the system may undergo some extreme scenarios, yielding severe contingencies that are not included in the contingency list, such as those *N-k* contingencies. In this case, the emergent controls have to be used, which are out of the scope of this article. It is also worth noting that once these unstudied scenarios happen, they can be easily included in the training set and a new model can be obtained so that it can deal with it next time. By iterating the process, the model will become better and better.

In this article, some high probability and severe contingencies determined by the operator according to engineering experiences are selected to verify the proposed method. For the three test

TABLE IX
COMPARISON OF THE TSI BEFORE AND AFTER PREVENTIVE CONTROL CONSIDERING *N*-2 CONTINGENCIES

N-2 Fault ID		39-bus system Before control	39-bus system After control	68-bus system Before control	68-bus system After control	140-bus system Before control	140-bus system After control
LS	1	**-99.8**	**71.6**	78.8	86.8	63.5	77.9
	2	**-99.8**	**71.7**	**-84.0**	**67.5**	**-99.6**	**75.4**
	3			**-92.6**	**86.8**	**-99.6**	**71.7**
	4					**-99.8**	**69.4**
	5					69.9	76.7
GO	1	**-99.4**	**72.3**	79.6	86.5	**-99.8**	**72.3**
	2	74.5	72.7	**-98.6**	**75.0**	**-99.6**	**78.8**
	3	**-99.5**	**70.3**	77.7	86.8	**-99.8**	**66.7**
	4			**-89.2**	**54.4**	**-99.6**	**77.0**
	5			75.1	74.1	**-99.6**	**78.1**
	6					63.0	77.6
	7					63.2	78.1
	8					67.8	80.1
	9					65.5	75.7
	10					**-94.9**	**65.7**

LS: line switching; GO: generation outage.
The significance of bold entities is to highlight numbers, which successfully achieves preventive control.

TABLE X
COMPARISON OF THE METHOD IN THIS ARTICLE AND THE METHOD IN [13]

Objectives	Method Proposed	Method [13]
TSI	60	60
Total cost	1146	2361
Generators load ratio	62.4	63.1
Transmission lines load ratio	44.4	46.7

systems, we have added 2, 3, and 5 *N*-2 line switching contingencies and 3, 5, and 10 *N*-2 generation outage contingencies, respectively. These cases are integrated with the previous samples to retrain the DBN. The TSI comparison results of *N*-2 contingencies before and after the preventive control are shown in Table IX. We can see that all *N*-2 contingencies are stable after preventive control, which proves that the proposed method is suitable for *N*-*k* contingencies and topology changes.

G. Comparison With Other Models

In [13], a transient stability preventive control model based on ANN and imperialist competitive algorithm (ICA) is proposed. The proposed method is compared with it on the 39-bus system, and the results are shown in Table X. Note that the same initial scenarios are selected and the ANN[4] is embedded in ICA. ICA selects the same parameters as the NSGA-III algorithm, and only generator redispatch is considered. Since ICA can only solve the single-objective optimization problem, the weighted sum of multiobjective is taken as the optimization objective of ICA. It can be seen from Table X that the preventive control results by our proposed method are much better than the single-objective evolutionary algorithm in [13]; see the total cost and generator load ratio.

V. CONCLUSION

In this article, a new computing paradigm that integrates deep learning-enabled surrogate model and the multiobjective evolutionary algorithm was proposed for power system transient stability preventive control. The key idea was to strategically integrate the DBN-enabled surrogate model with the NSGA-III algorithm when solving the optimal preventive control strategy with various equality and inequality constraints. DBN allows users to develop the surrogate model for system transient stability assessment, different from the traditional time-consuming time-domain simulations. The surrogate model significantly reduced the computational complexity of the full TSC-OPF by simply performing direct deep learning-based transient stability verification in each iterative searching step for the optimal preventive control strategy. Comparative results on several IEEE test systems showed that the proposed method achieved significantly improved computational efficiency without loss of control performance. In our future work, the developed method will be tested using practical systems and data.

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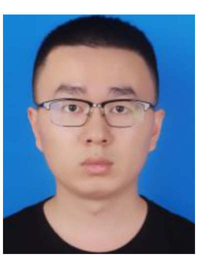

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